



Indian Institute of Technology, Madras

EDUCATION			
Program	Institution	CGPA/%	Year
BTech in Chemical Engineering <i>Minor: Energy and Environment*</i>	Indian Institute of Technology, Madras		2021
Class XII (CBSE)	Delhi Public School, Mumbai		2017
Class X (CBSE)	Ryan International School, Mumbai		2015
PROFESSIONAL EXPERIENCE			
Dr. Reddy's Laboratories <i>Intern</i> (May-July 2020)	<i>Mapping facility and assessing the market for compounds (Ligands, Catalysts and mPEGs¹)</i> - Built and assembled Process Engineering packages containing Process Flow Diagrams, Process and Instrumentation diagrams, Material Safety Data Sheet and Raw Material Cost Sheet of four major products (2 ligands, 1 catalyst and 1 mPEG¹) that formed the backbone for mapping the facility in India - Identified four critical bottlenecks in the formation process of compounds and suggested modifications for the same; implementation of those modifications during the formation process will lead to increase in the final gain and purity in the synthesis of Ligands, Catalysts and Methoxy Polyethylene Glycols - Analyzed the market for the compounds; identified competition from all around the world (50 major competitors), knowledge of the competition will help in identifying the next business moves - Performed detailed volume and value analysis for the compounds, which further helped in assessing different marketing strategies that are taken by different competitors and their production costs - Reported customers served in the past and determined whether those relationships can be revived - Identified a new key application of mPEGs¹ in the construction industry as a super plasticizer that has a significant market share of 30% in terms of consumption, production and revenue; branching into the identified field with plant scale synthesis of mPEGs¹ will bring in huge profits for the company - Scaled up the formation process of all the compounds from lab to plant scale wherever required		
Hindustan Petroleum Corporation Limited <i>Intern</i> (May-June 2019)	<i>Basic understanding of the crude unit and mitigation measures to revamp the existing crude unit</i> Analyzed the steps undertaken to revamp the existing crude unit by examining the Process Flow diagrams Compared existing and post revamp fuel refining units and identified major equipment changes by analyzing the Process and Instrumentation diagrams and on-site field evaluation - Calculated the NPSH² value of pump by thoroughly examining the Process and Instrumentation diagrams		
EXTRA CURRICULAR ACTIVITIES			
Sports	- Part of Institute Badminton Team B (2018-present) - Represented IIT Madras in Badminton Sports-Fest (Inter College Sports Tournament) consecutively for 2 years, secured position in Team A in the year 2019 by winning fourth position in the year 2018 - Represented the hostel in Badminton Sports Schroeter (Inter Hostel Sports Tournament) consecutively for 2 years, won fourth position in the year 2019 and improved it to third position in the year 2020 Won third position in 400m relay Athletics Schroeter (Inter Hostel Sports Tournament, Sept 2019) - Selected in 20-member National Sports Organization team out of a batch of 900 (2017-2018) Sharavati Hostel Sports: Volunteer (2019-2020) - Devised ways of increasing participation in sport events and successfully managed to achieve an increase of 10 percent; publicized events on social media and motivated fitness among residents of the hostel - Organized, participated and managed logistics in three mixed tournaments (Badminton, TT and Foosball) - Conducted intra-hostel sport events like dodgeball to reduce stress among residents during exams		
Pathshala <i>Volunteer</i> (Nov-Dec 2018)	- Guided a group of 20 underprivileged students in academics and overall personality development Identified their weaknesses and strengths; devised ways to enhance their communication skills - Collaborated with other volunteers and organized two workshops that allowed students the opportunity to showcase their talents and skills, which further helped in boosting their confidence - Initiated a scoreboard system based on assignments and pop quizzes which contained their present score and their desired score for next week to bring out the competitive spirit among the children Collected feedback on a weekly basis from both the student as well as the parent to do better each week; outlined and achieved goals set in the starting of the week		
LitSoc (2018-2020)	Participated consecutively for two years in LitSoc (Inter Hostel Cultural Competition); handled the logistics behind events and increased the turn-out of participants from the hostel by 15 percent each year		
Saathi ³ Mentor (2018-2019)	Counselled and assisted a group of four freshman students in coping with change in environment and opened them up to co-curricular and extra-curricular activities offered at IIT Madras		
COURSEWORK AND SKILLS			
Renewable Energy Sources	Fundamentals of Operations Research	Applications of Heat and Mass Transfer	
Nuclear Energy Option*	Operations Management	Fluid and Particle Mechanics	
Unconventional Oil and Gas Resources	Integer Optimization	Continuum Mechanics & Transport Phenomenon	
Process Analysis and Simulation*	Principles of Economics	Computational Programming	
Process and Product Design*	Applied Economics*	Differential Equations	
Software: MATLAB, Aspen Plus, Simulink		Programming Languages: C, C++, SQL	
Languages: Hindi, English, Puniabi, Sanskrit, German, Chinese			